

HST/JWST Science Justification Proposal Structure

HST Programs

GO Programs:

Very Small 1-15 orbits (relatively new)
Small 16-34 orbits
Medium 35-74 orbits
Large/Treasury > 75 orbits

Archival Programs:

- Regular (??~\$50K ; < \$120K ???)
- Legacy (??> \$120 K??)
- Theory (??~\$50K ; < \$120K??)
- Cloud Computing (??)
- AR Data Science Software

<https://hst-docs.stsci.edu/hsp/hubble-space-telescope-call-for-proposals-for-cycle-34/hst-proposal-categories>

JWST Programs

GO Programs:

Very Small < 20 hrs
Small 20-50 hrs
Medium 50-130 hrs
Large/Treasury > 130 hrs

Archival Programs:

- Regular (????)
- Legacy (??)
- Theory (??)
- Cloud Computing (??)
- Data Science Software

<https://jwst-docs.stsci.edu/jwst-opportunities-and-policies/jwst-call-for-proposals-for-cycle-5/jwst-proposal-categories#gsc.tab=0>

Required Components

- **1) Abstract**
- **2) Science Justification**
- **3) Technical Details**
 - GO: Technical Justification
 - AR: Analysis Plan

HST/JWST page limits proposal

-Page limits for
Sci Just +
Technical Details

-No limits on
number of
figures, but total
page count
matters

-12 pt font

-References not
included in page
count

Proposal Category ¹	Total Page Limit for the Required Sections in the PDF Attachment ²
Very Small GO Small GO Regular AR Regular AR/Theory Snapshot	4 THIS CLASS Science Justification should be ~ 2.5 pages
Very Small GO-AR Small GO-AR	5
Medium GO Medium GO-AR	5
Large GO Large GO-AR Treasury GO	6

<https://hst-docs.stsci.edu/hsp/hubble-space-telescope-call-for-proposals-for-cycle-34/hst-preparation-of-the-pdf-attachment#HSTPreparationofthePDFAttachment-scientificjustificationScientificJustification>

<https://jwst-docs.stsci.edu/jwst-opportunities-and-policies/jwst-call-for-proposals-for-cycle-5/jwst-preparation-of-the-pdf-attachment#JWSTPreparationofthePDFAttachment-pagelimitsPagelimits>

HST Sci Justification Instructions

Scientific Justification

This section should present a balanced discussion of background information, the program's goals, its significance to astronomy in general, and its importance for the specific sub-field of astronomy it addresses. The members of the [review panels](#) will span a range of scientific expertise, so you should write this section for a general audience of scientists.

Depending on the type of proposal, the following items should also be included:

- GO Treasury, AR Legacy, and Pure Parallel Proposals should address the value to the astronomical community of the data products that will be generated by the program.
- Proposals using ACS/WFC, WFC3/UVIS, or WFC3/IR for undithered imaging must explain why this strategy is needed for the scientific objectives; dithering is required to eliminate hot pixels and other detector artifacts that may compromise the archival value of the data.
- ACS/SBC, COS, and STIS/MAMA proposers must address the safety of their targets and fields with respect to the appropriate count rate limits of the photon-counting detectors (see Chapter 5 of the Primer and the COS, STIS, or ACS Instrument Handbooks).
- SNAP Proposals should provide a complete description of the target sample.
- AR Proposals should describe how the project improves upon or adds to the previous use of the data.
- Theory Proposals should include a description of the scientific investigation that will be enabled by the successful completion of the program, and their relevance to HST.
- Calibration AR Proposals should describe what science will be enabled by the successful completion of the program, and how the currently supported core capabilities, their calibrations, and the existing pipeline or data reduction software are insufficient to meet the requirements of this type of science.

JWST Sci Justification Instructions

Proposal narrative

The PDF attachment must contain a Proposal Narrative with sections that discuss the following topics.

1. Scientific Justification: This section should present a balanced discussion of background information, the program's goals, its significance to astronomy in general, and its importance to for the specific sub-field of astronomy it addresses. The members of the review panels will span a range of science expertise, so one should write this section for a general audience. Depending on the type of proposal, the following items should also be included:

- Treasury GO, Legacy AR, and Pure Parallel proposals should address the value to the astronomical community of the data products that will be generated by the program.
- Survey proposals should provide a complete description of the target sample.
- AR proposals should describe how the project improves upon or adds to the previous use of data.
- Theory proposals should include a description of the scientific investigations that will be enabled by the successful completion of the program, and their relevance to JWST.
- Calibration proposals should describe what science will be enabled by the successful completion of the program, and how the currently supported core capabilities, their calibrations, and the existing data processing are insufficient to meet the requirements of this type of science.
- Community Data Science Software Proposals should describe how the software packages that will be developed are relevant to and necessary for the reduction or interpretation of JWST data.

Abstract Outline

Facts

Importance of Facts

Potential Impact

Problem

This sets the title and the narrative

Goal – identify the “key component” that will solve the problem using a “target”

We propose to...

Strategy – to utilize/generate the key component , (Justify HST/JWST)

Suitability

Importance of Solution

Impact within Sub Field

Broader Impact

Out-of-field Impact – also informs the narrative

Outline for Science Justification

Abstract: Summarizes the entire proposal

Section 1: Facts and summary of Sections 2-4

Section 2: The Problem/Motivation/Story

Section 3: The Goal & Strategy. How you will use “key component” + “Target” to solve the Problem

Section 4: Importance of the solution to the subfield & Broader Impacts (legacy value)

New Models of the Milky Way's Dark Matter Distribution for the Era of High Precision Astrometry

Proposal 1: HST AR Cycle 26 PI Besla

Facts

Understanding the assembly history and dark matter distribution of our Milky Way (MW) is a major challenge for astrophysics. Thanks to the unique capabilities of HST, proper motions of satellite galaxies, globular clusters and stellar streams have been measured with accuracies of order ~ 0.05 mas/yr (~ 10 km/s) at distances

Importance

of 50-300 kpc. When combined with detailed models of the MW's halo potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW and compute accurate orbital histories of satellites.

Problem

However, the MW hosts a pair of massive dwarf galaxies, the LMC and SMC, that contribute to its dark matter distribution and change the shape of the potential in a non-symmetrical, time evolving manner. To date, these effects have not been accounted for in existing models of the MW halo.

Goal

We propose to develop high resolution simulations to quantify the time evolving structure of the MW's dark matter halo owing to the influence of the LMC and SMC.

Strategy

These novel models will enable rapid orbital integration of halo objects (satellites, globular clusters, stellar streams), using high accuracy HST proper motions, while also capturing the complex halo potential resulting from the LMC-SMC-MW interaction.

The era of high-precision astrometry has arrived, yet we do not currently have an appropriate theoretical framework to study the assembly history of MW-like galaxies in the presence of massive satellite perturbers.

Our proposed program is thus critical to ongoing HST programs and all efforts to understand the structure and evolution of the dark matter halo of our Galaxy and analogous systems like M31 and its massive satellite, M33.

Importance of solution

Broader Impacts

Science Justification: Section 1

The Big Picture

- Use a descriptive heading (not “intro” / “backgrd”)
- **1-2 paragraphs:**
 - Explain/Justify “the Facts” stated in abstract
 - Highlight role of observatory in establishing facts.
 - Explain the importance of “the Facts”
- **1 paragraph:** Introduce the “Problem” (to be expanded – Section 2)
- **Synopsis: Last paragraph:** (to be expanded - Section 3-4)
 - State the GOAL
 - State that this program will fix “the problem” by doing *blah*
 - Big picture statement to end it off

What is a Paragraph?

- Every paragraph should have exactly one topic that is distinct from the topics of other paragraphs.
- Topic is highlighted in the FIRST sentence. Followed by information that clarifies the main topic.
- Every paragraph should have a purpose

From sentence to paragraph: topic strings

Mixed topic strings (combine these two):

Character 1

Character 1

Character 1 is related to character 2. Character 2

Character 2

Character 2 introduces Character 3

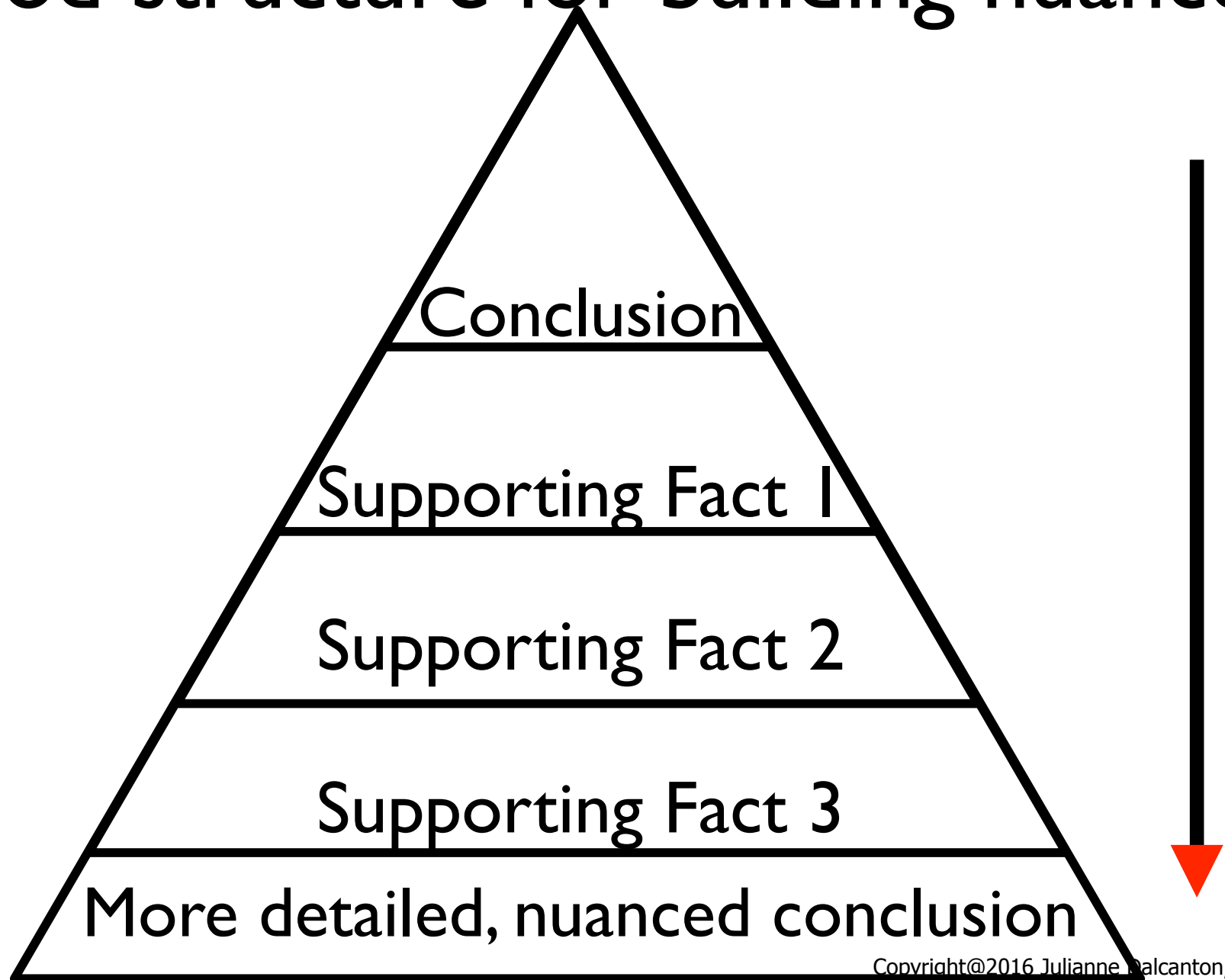
Character 3...

Concluding sentence that recaps the relationship between Character 1, 2, and 3, or collapses the middle term and posits a direct relationship between character 1 and character 3

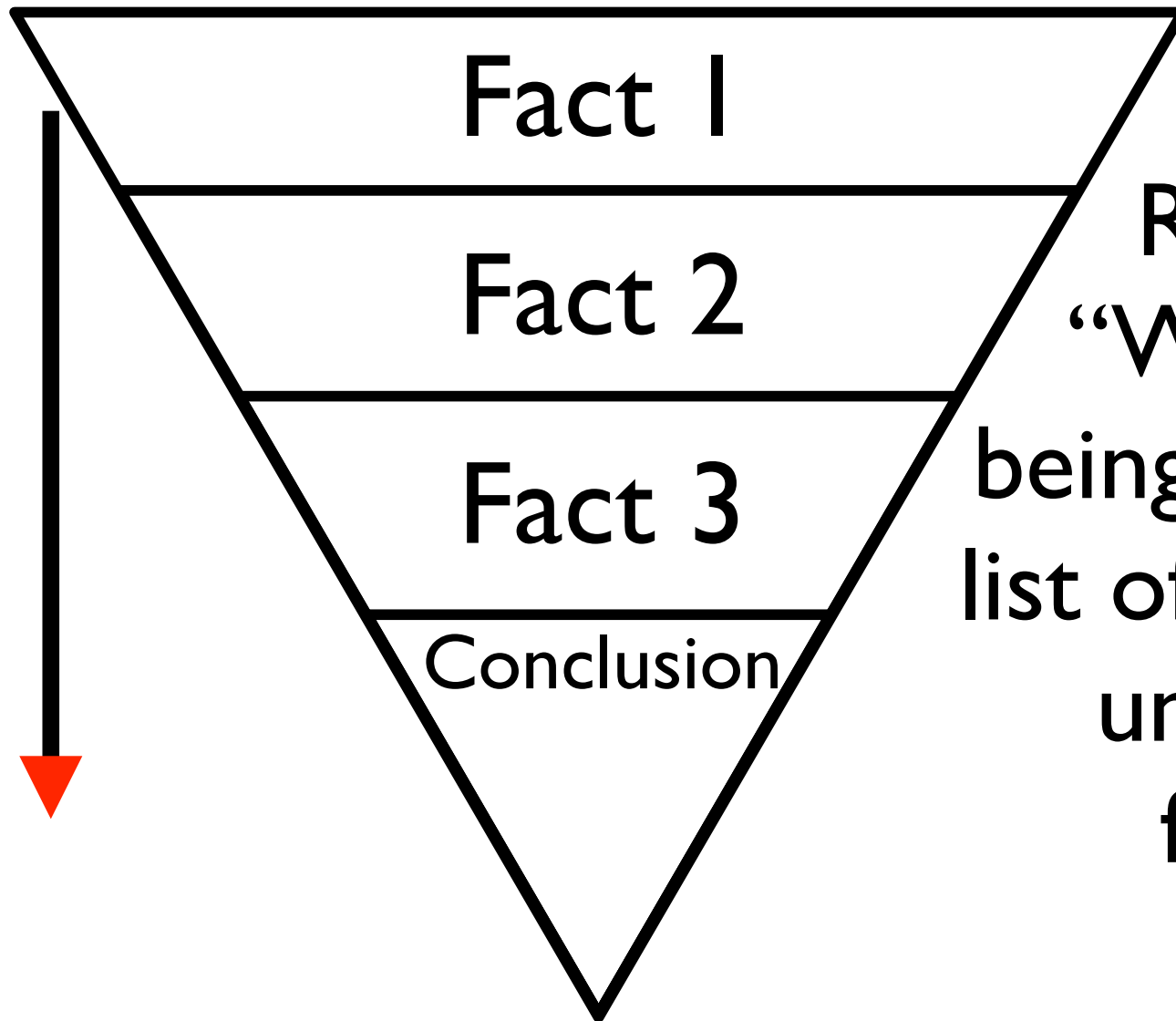
There shouldn't be more than 2-3 characters !!

Dr. Steph Brown
UA English Department

Good structure for building nuance.



Contrast with



Reader:
“Why am I
being given this
list of seemingly
unrelated
facts?”

Only include information crucial to the narrative

- You should *minimize and/or leave out* selling points that distract from your story.
- The reader (reviewer) may believe that *anything you mention* is part of your main story.
- They will then mark you down for not fully justifying something you considered to be a minor aside.

Facts/Importance from Abstract

Understanding the assembly history and dark matter distribution of our Milky Way (MW) is a major challenge for astrophysics. Thanks to the unique capabilities of HST, proper motions of satellite galaxies, globular clusters and stellar streams have been measured with accuracies of order ~ 0.05 mas/yr (~ 10 km/s) at distances of 50-300 kpc. When combined with detailed models of the MW's halo potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW and compute accurate orbital histories

Title: Argument being made using facts

The Need for New Milky Way Halo Models and the Role of the Magellanic Clouds

Astronomy has entered an era of high precision astrometry. Data from the *Hubble Space Telescope* (HST), the *James Webb Space Telescope* (JWST), and survey missions, such as *Gaia* and *LSST* have/will be used to identify substructure about the Milky Way (MW) and measure their kinematics with unprecedented accuracy. For example, the recently approved HST Large Cycle 24 program GO-14734 (PI Kallivayalil; Co-I Besla) will result in first epoch imaging to lay the foundation for proper motion measurements for *all* satellite galaxies of the MW (precision of ~ 0.05 mas/yr, or ~ 10 km/s). High accuracy, 6-D phase space information of halo tracers (satellites, globular clusters, stellar streams) over a wide range of Galactocentric distances is thus imminent. When combined with detailed models of the MW's halo potential, such measurements become high-precision tools to constrain the dark matter mass distribution about the MW and compute accurate orbital histories of satellites/streams, revealing the MW's assembly history.

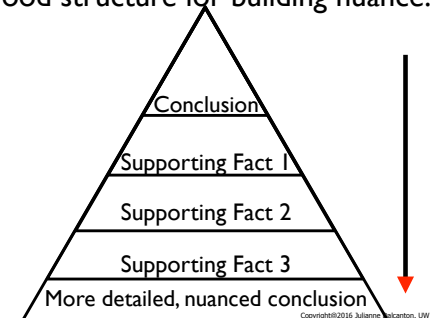
Conclusion

Supporting Fact 1 – role of HST/JWST

Supporting Fact 2 – Specific role of HST

More Detailed, Nuanced Conclusion – leads to next paragraph

Good structure for building nuance.



Problem from Abstract

of satellites. However, the MW hosts a pair of massive dwarf galaxies, the LMC and SMC, that contribute to its dark matter distribution and change the shape of the potential in a non-symmetrical, time evolving manner. To date, these effects have not been accounted for in existing models of the MW halo. We propose to develop high

There is, however, a problem with all existing MW models. HST proper motions imply that the MW has recently captured a pair of massive, high-speed satellites: the Large Magellanic Cloud (LMC) and Small Magellanic Cloud (SMC) (Besla+2007; Kallivayalil, van der Marel, Besla+2013). These galaxies are cosmologically expected to

contribute $\sim 2 \times 10^{11} M_{\odot}$ of dark matter to the MW halo (Besla+2012,2010). This is roughly $\sim 20\%$ of the total expected mass of the MW (Boylan-Kolchin, Besla+2011).

Thus, the Clouds are not small galaxies. Indeed, the stellar disk of the LMC has a radius that is $\sim$ half its current separation to the MW!! (Besla+2016; Fig. 1). As such, the Clouds

cannot be treated as simple point mass tracers of the halo potential – rather, such massive satellites will *contribute* to the dark matter distribution of the MW, changing the shape of the potential in a non-symmetrical, time evolving manner. This effect has not yet been

accounted for in existing models of the MW, but has the potential to perturb the kinematics of *all* tracers of the halo potential.

Conclusion

Supporting Fact 1

Supporting Fact 2

Supporting Fact 3

**More Detailed, Nuanced Conclusion
→ Leads into the goal statement**

Synopsis: Last Par. of Section 1

We propose to create needed N-body models of the LMC+SMC+MW system with mass resolution 3 orders of magnitude higher than any created to date (10^9 particles). By applying a basis expansion method, we will extract the shape, density and potential of the dark matter halo in the simulated combined MW+LMC+SMC system as a function of time and radius. These simulations will be the *first of their kind*, accounting for our new understanding of both the large mass and HST proper motions of the Clouds. Our solutions for the time evolving potential of the MW will be released to the community, providing necessary tools to interpret existing and upcoming astrometric data sets from HST, JWST and *Gaia*, and facilitate the orbital analysis of new substructure as it is discovered. This program is thus both critical and timely.

Goal/Proposal : Key Component (HST PM), Target (Bridge) – to be expanded upon

The plan to use Models to fix the Problem – to be expanded upon

Importance of solution to subfield and role of HST - to be expanded upon

Broader Impact / Community Products / HST JWST connection - to be expanded upon

Section 1 sets up 3 things to be expanded upon:

- Section 2 : The Problem/Narrative – leads to the key component (HST/JWST/Models) needed to solve the problem
- Section 3: The Strategy – Demonstrate HST/JWST/Models (key component) + “Target” will solve the problem
- Section 4: Importance to the Subfield & more broadly (Legacy Value) + HST/JWST relevance

A S**P**ectroscopic survey of biased halos In the Reionization Era (ASPIRE): A JWST Quasar Legacy Survey

After two decades of search, the first large sample of quasars has been identified in the reionization era. We propose to obtain NIRC*am* observations of a flux-limited sample of 25 quasars at $6.5 < z \leq 6.8$ with extant high resolution ALMA sub-mm observations and deep optical-to-infrared spectra. This program will enable a powerful spectroscopic (WFSS mode) and imaging survey along the entire quasar light cones, resulting in the detection of H*beta*+*[OIII]* lines of ~ 350 galaxies at $5.3 < z < 7$ over 240 arcmin^2 sky area, including 45 galaxies physically associated to the central quasars. It will finally resolve the long-standing question of whether the earliest supermassive black holes (SMBHs) reside in the most massive dark matter halos and inhabit large scale galaxy overdensities. We will simultaneously image the host galaxies and close companions of quasars, measure the masses of the central SMBHs and characterize quasar feedback with H*beta*+*[OIII]* emissions, providing unprecedented constraints on the connection between SMBHs and their hosts as well as their primordial environment. This program will also provide unparalleled constraints on cosmic reionization and galaxy formation by measuring ionizing photon escape fractions of faint galaxies and probing the circumgalactic media of galaxies at $z \sim 5-7$. In addition, this survey will give the most accurate bright-end galaxy luminosity function and H*beta*+*[OIII]* equivalent width measurements at $z \sim 5-7$, complementary to the GTO JADES deep spectroscopic survey. Finally, the coordinated NIRISS parallel imaging will allow us to identify additional galaxies at $z \sim 6.5-6.8$ to probe quasar-galaxy clustering at larger scales.

Facts, no importance

Problem

Goal

JWST Strategy

Importance of solution

Broader Impacts – This is a Medium Program (61.5hrs)

Science Justification - FACTS

From abstract

After two decades of search, the first large sample of quasars has been identified in the reionization era.

Since the discovery of the first $z > 6$ quasar (Fan et al., 2001), over two decades of painstaking searches have delivered a sample of ~ 50 of these extremely rare objects at $z > 6.5$ up to the current redshift record at $z = 7.5$ (Bañados et al., 2018). The existence of these quasars powered by $\sim 10^9 M_{\odot}$ black holes (e.g. Mortlock et al., 2011; Yang et al., 2020a) in the Epoch of Reionization (EoR) challenges our understanding of supermassive black hole (SMBH) formation. An assortment of cosmological simulation models are able to produce

Conclusion

Supporting Facts

More Detailed Conclusion

Science Justification - FACTS

From abstract

After two decades of search, the first large sample of quasars has been identified in the reionization era.

(SMBH) formation. An assortment of cosmological simulation models are able to produce such massive SMBHs (e.g. Di Matteo et al., 2005; Khandai et al., 2015) powering quasars with roughly the observed abundance 0.4 Gpc^{-3} (Wang et al., 2019) starting with massive $\gtrsim 10^4 M_{\odot}$ seed BHs. These models generically predict that these SMBHs are hosted by massive $M_{\star} \gtrsim 10^{11} M_{\odot}$ galaxies with star formation rate $\text{SFR} \gtrsim 100 M_{\odot} \text{ yr}^{-1}$, and reside in the rarest $M_{\text{DM}} \gtrsim 10^{12.5} M_{\odot}$ halos situated in the most overdense regions (Costa et al.,

Conclusion 2

Supporting Facts

More Detailed Conclusion

Science Justification – Problem Statement

From abstract

physically associated to the central quasars. It will finally resolve the long-standing question of whether the earliest supermassive black holes (SMBHs) reside in the most massive dark matter halos and inhabit large scale galaxy overdensities. We will simultaneously image the host galaxies and close companions of quasars,

2014). While these numerical studies have established the plausibility of the existence of $z \sim 7$ quasars in a cosmological context, to date, testing these theories has proven impossible given the current sensitivity of observations. As we argue in this proposal, JWST will deliver the sensitivity to finally test these predictions in detail, answering long-standing questions about the primordial environment and the fueling mechanisms of the earliest SMBHs.

Conclusion – problem statement

Supporting Facts

More Detailed Conclusion – we have the solution

Synopsis: Last Par. of Section 1

We propose to conduct a spectroscopic and imaging survey of **25** quasars in the EoR with NIRC*am*. This survey will enable a powerful *redshift survey* along the entire quasar light cones, resulting in the detection of $H\beta + [O\ III]$ lines of $\sim$ **350** galaxies at $5.3 < z < 7$ over $\sim$ **240** arcmin² of the sky. It will also image the host galaxies and close companions of quasars spanning the rest-frame UV to optical. This program will allow us to:

- Measure the distribution of star-forming galaxies around these quasars to finally resolve whether the earliest SMBHs reside in the most massive dark matter halos.
- Detect or tightly constrain the UV-optical continuum emitted by stars in quasar hosts, providing unprecedented constraints on the galaxies hosting these SMBHs.
- Precisely measure the mass of the SMBHs with the $H\beta$ line width and characterize whether quasar feedback regulates the formation of the earliest SMBH-galaxy systems.
- Assemble a large statistical sample of foreground galaxies along quasar light cones probed by co-spatial absorption spectroscopy allowing us to 1) constrain ionizing photon escape and the galaxy contribution to cosmic reionization and 2) probe the metal enrichment and ionization state of circumgalactic medium of galaxies at $z \sim 5 - 7$.

Goal/Proposal

The plan to use HST to fix the Problem

Solution – for Subfield

Broader Impact – Bonus Science

Dual Anonymous

[Examples](#)

Guidelines for proposers

This section provides guidelines to assist proposers prepare their proposals, specifically their PDF Submissions, to avoid highlighting their identity and to support a fairer proposal evaluation process. The anonymous review does not mean proposals will be accepted from anonymous sources. As with previous cycles, proposers must still enter the names and affiliations of all investigators into the APT system.

APT will not include names or affiliations in the versions generated for the reviews.

While APT will largely obscure the proposing teams identities in cover materials, it will not change or alter information contained in the PDF submission. **Thus, it is necessary for proposers to take additional steps to further anonymize their PDF attachment before it is uploaded to APT.** Below are some guidelines to accomplish this:

- Do not include author names or affiliations anywhere in the PDF attachment. This includes but is not limited to, page headers, footers, diagrams, figures, or watermarks. This does not include references to past work, which should be included whenever relevant (see below).
- Referencing is an essential part of demonstrating knowledge of the field and progress. When citing references within the proposal, use third person neutral wording. **This especially applies to self-referencing.** For example, replace phrases like “as we have shown in our previous work (Doe et al. 2010)” with “as Doe et al. (2010) showed...” Do not refer to previous campaigns using HST or other observatories in an identifying fashion. For instance, rather than write “we observed another cluster, similar to the one we are proposing under HST program #XXXXX,” instead write “HST program #XXXXX has observed this target in the past...”
- We encourage references to published work, including work citable by a DOI. It may be occasionally important to cite exclusive access datasets or non-public software that may reveal (or strongly imply) the investigators on the proposal. We suggest proposers use language like “obtained in private communication” or “from private consultation” when referring to such potentially revealing work.
- Do not include acknowledgements, or the source of any grant funding.

It takes some effort by authors to anonymize their PDF submissions. As the guidelines show, grammar and structure are expected to be different than in many other submissions of this type, including previous HST submissions. Some examples of re-worked text can be found on [Example text for anonymous proposing](#). **Proposers should take sufficient time to prepare the manuscript, especially if one plans to rework a proposal from other submissions.**

